WORKED SOLUTION

STEP III 2021, Question 10

Mechanics • Wrapping String Motion

STEP III 2021, Question 10

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

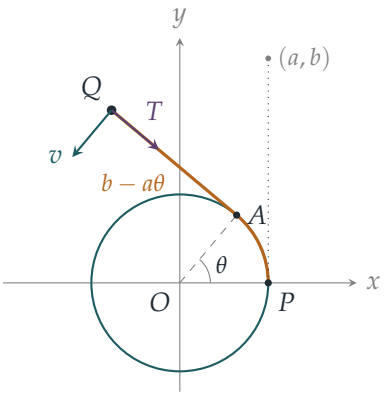
The key idea

Describe everything by the angle θ through which the straight part of the string has turned. The string leaves the cylinder along a tangent, so the particle sits at the point of tangency plus a tangent vector of length $b - a\theta$. Differentiating shows that the particle's velocity is always at right angles to the string, so the tension does no work and the speed never changes. At each instant the particle is moving on a circle centred at the point where the string leaves the cylinder, and the tension supplies the centripetal force.

Solution

We model the particle as a point moving on a smooth horizontal table, so the only horizontal force on it is the tension T in the light inextensible string (the weight is balanced by the normal reaction of the table). The string is taut throughout.

For the string to wrap round the cylinder, the impulse must send the particle in the negative x-direction from (a, b), so the string turns anticlockwise. Let A be the point where the straight part of the string leaves the cylinder at time t.



Part (i)

The point A is $(a\cos\theta, a\sin\theta)$, since the radius to P has turned through θ. The length of string wrapped on the cylinder is the arc PA, of length $a\theta$, so the straight part has length $b - a\theta$. It lies along the tangent at A, in the direction of increasing θ, which is the unit vector $(-\sin\theta, \cos\theta)$. (At $\theta = 0$ this is $(0, 1)$, matching the initial position (a, b).) So the particle is at

$$x = a\cos\theta - (b - a\theta)\sin\theta, \qquad y = a\sin\theta + (b - a\theta)\cos\theta.$$

Differentiating with respect to t, using the product rule and $\frac{\mathrm{d}}{\mathrm{d}t}(b - a\theta) = -a\dot{\theta}$,

$$\dot{x} = -a\dot{\theta}\sin\theta + a\dot{\theta}\sin\theta - (b - a\theta)\dot{\theta}\cos\theta = -(b - a\theta)\dot{\theta}\cos\theta,$$
$$\dot{y} = a\dot{\theta}\cos\theta - a\dot{\theta}\cos\theta - (b - a\theta)\dot{\theta}\sin\theta = -(b - a\theta)\dot{\theta}\sin\theta.$$

Hence $\dot{x}^2 + \dot{y}^2 = (b - a\theta)^2\dot{\theta}^2$, and since $b - a\theta > 0$ and $\dot{\theta} > 0$ (the string is wrapping on), the speed is

$$v = (b - a\theta)\dot{\theta}.$$

Part (ii)

Ask yourself Why should the speed stay equal to u? The string is pulling on the particle the whole time.

From part (i) the velocity is $-(b - a\theta)\dot{\theta}\,(\cos\theta, \sin\theta)$, which is parallel to the radius OA and therefore perpendicular to the string, whose direction is $(-\sin\theta, \cos\theta)$. The tension acts along the string, so its rate of doing work, $\mathbf{T}\cdot\mathbf{v}$, is zero at every instant. No other horizontal force acts, so the kinetic energy, and hence the speed, is constant. Initially the speed is u, so

$$\dot{\theta}\,(b - a\theta) = u,$$

as required.

Part (iii)

Using (ii), the velocity is $\mathbf{v} = -u(\cos\theta, \sin\theta)$ with u constant, so the acceleration is

$$\dot{\mathbf{v}} = -u\dot{\theta}(-\sin\theta, \cos\theta).$$

This has magnitude $u\dot{\theta}$ and points from the particle towards A, exactly along the string. (Equivalently: at each instant the particle is moving on a circle of radius $b - a\theta$ about A at speed u, so its acceleration is $u^2/(b - a\theta)$ towards A.) Newton's second law along the string gives

$$T = mu\dot{\theta} = \frac{mu^2}{b - a\theta}.$$

It remains to express $b - a\theta$ in terms of t. Integrating (ii) with respect to t, using $\theta = 0$ at $t = 0$,

$$b\theta - \tfrac{1}{2}a\theta^2 = ut.$$

Then

$$(b - a\theta)^2 = b^2 - 2a\left(b\theta - \tfrac{1}{2}a\theta^2\right) = b^2 - 2aut,$$

and since $b - a\theta > 0$ we take the positive root, $b - a\theta = \sqrt{b^2 - 2aut}$. Therefore

$$T = \frac{mu^2}{\sqrt{b^2 - 2aut}},$$

as required.

Checks

- At $t = 0$ the tension is mu^2/b: the particle starts moving on a circle of radius b about P at speed u, as expected.
- The string is completely wound when $a\theta = b$, which happens at $t = b^2/(2au)$; then $b^2 - 2aut = 0$ and the tension grows without bound, as the radius of the instantaneous circle shrinks to zero at constant speed.
- Dimensions: b^2 and aut are both areas, and mu^2/length is a force.
- We confirmed the results by computer, stepping the motion numerically for several random choices of a, b, u and m and differentiating the position numerically: the speed stayed at u and the measured force along the string matched $mu^2/\sqrt{b^2 - 2aut}$ to about 3×10^{-5} (relative).

Common slips

- Assuming angular momentum about O is conserved. It isn't: the tension acts along a tangent to the cylinder, which does not pass through O, so it has a moment about O. Energy is the conserved quantity here.
- Using the length b instead of $b - a\theta$ for the straight part, forgetting the string already wrapped on the cylinder.
- Taking the radius OQ of the particle's path, rather than the straight length AQ, as the radius of curvature in the tension calculation.

Beyond STEP

The path traced by the end of a string unwinding from (or winding onto) a curve is called an *involute*; the involute of a circle is the shape of most gear-tooth profiles, because it makes meshing gears transmit motion smoothly. The question also illustrates a general principle met in first-year mechanics: a constraint force perpendicular to the velocity does no work, so energy can be conserved even when momentum and angular momentum are not. Kleppner and Kolenkow, *An Introduction to Mechanics*, has a good discussion of exactly this kind of problem.

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